

Aditya Sinha

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EDUCATION

B.Tech, Computer Science and Engineering

2023 – 2027

VIT Vellore (CGPA: 8.52)

Senior Secondary (Class XII, TSBIE)

2023

Sri Chaitanya Junior Kalasala, Hyderabad

Secondary (Class X, CBSE)

2021

Delhi Public School, Hyderabad

EXPERIENCE

AI/ML Intern — Missile Telemetry Data Analysis

[May, 2026 – July, 2026]

Defence Research & Development Laboratory (DRDL), DRDO | Kanchanbagh, Hyderabad, India

Python, pandas, NumPy, SciPy, Streamlit, Plotly, Time-Series Analysis, Anomaly Detection, EDA

- Built an automated Python (pandas/NumPy) pipeline to parse, clean, and filter high-volume raw missile telemetry, converting unstructured multi-channel sensor streams into structured, analysis-ready time-series datasets.
- Developed an unsupervised anomaly-detection tool (Z-score and IQR statistical methods) over multi-channel time-series telemetry to auto flag outliers and irregular signal patterns, cutting manual inspection effort.
- Built an interactive Streamlit and Plotly dashboard for exploratory data analysis and visual anomaly review by selecting parameters, overlaying flagged anomalies on time-series charts, and exporting results to CSV.

PROJECTS

[\[Link\]](#) ATS Resume Analyzer & Optimizer — Full-Stack NLP App

Python, FastAPI, Next.js, TypeScript, spaCy, Sentence-Transformers, scikit-learn, SQLite

- Built a full-stack résumé-screening app (FastAPI + Next.js/TypeScript) with 40 REST endpoints and a weighted 5-dimension ATS scoring engine matching resumes to any job description.
- Engineered the NLP core using spaCy, scikit-learn TF-IDF, rapidfuzz fuzzy matching, and all-MiniLM-L6-v2 sentence embeddings (cosine similarity) for semantic resume-to-JD matching and running fully local with no paid APIs.
- Backed the engine with a 240-test pytest suite across parsing, scoring, and API layers, with SQLite-persisted analysis history, bulk candidate ranking, and PDF/CSV report export via ReportLab.

[\[Link\]](#) Diabetic Retinopathy Classifier — Deep Learning

Python, TensorFlow/Keras, EfficientNetB4, Grad-CAM, scikit-learn, Streamlit

- Built a 5-class diabetic-retinopathy grading model via EfficientNetB4 transfer learning (ImageNet), with class-weighting and augmentation to handle a heavily imbalanced dataset (1,000 healthy to 190 severe-DR).
- Added Grad-CAM explainability to surface lesion regions per prediction, with a multi-model ensemble evaluator producing confusion matrices, per-class reports, and a Streamlit inference dashboard.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, SQL

Backend & APIs: FastAPI, Node.js, REST API design

Frontend: React.js, HTML/CSS, Streamlit

AI/ML: Scikit-learn, SHAP, FAISS, OpenAI API, NumPy, pandas, Matplotlib, SciPy

Databases: MySQL, MongoDB, Supabase (PostgreSQL)

Cloud & Tools: AWS (basics), Git, GitHub, Jupyter, Google Colab, VS Code

CS Fundamentals: Data Structures, Algorithms, OOP, DBMS, Operating Systems, Computer Networks

CERTIFICATIONS

- Model Context Protocol: Advanced Topics — **Anthropic**
- AI Fluency Framework Foundations — **Anthropic**
- Python for Data Science, AI & Development — **Coursera (IBM)**
- Generative AI using IBM watsonx — **IBM**
- Ultimate Fusion 360 Course — **Udemy**

ACHIEVEMENTS & LEADERSHIP

- Core Committee Member, roboVITics (robotics club), VIT — 2024–25
- National Space Hackathon – IIT Delhi — April 2025
- TALENT ARENA: More Than a Hack, Fira Barcelona — March 2026
- Student Ambassador, United Motorsport Academy — 2023